Muhammad Murtaza

AI Engineer (LLMs & Agentic AI)
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Summary

Computer Science graduate (GIKI, CGPA 3.54) building LLM-powered applications in Python: stateful agent workflows with LangGraph, RAG pipelines with ChromaDB, LoRA fine-tuning of Flan-T5, and async FastAPI backends with PostgreSQL. One of about 1,100 teams invited to present on-site at the Alibaba Cloud AI Hackathon Pakistan 2026.

Education

Ghulam Ishaq Khan Institute of Engineering Sciences and Technology (GIKI) Pakistan
B.S. Computer Science, CGPA: 3.54/4.0, Dean's Honor Roll (5 semesters) *Sep. 2022 – Jun. 2026*
- Relevant Coursework: Machine Learning, Deep Learning, Generative AI, Data Structures & Algorithms, Object Oriented Programming, DevOps, Design & Analysis of Algorithms

Experience

Software Engineering Intern Jul. 2025 – Sep. 2025
Techlogix, Client: SECP Pakistan *Islamabad, Pakistan*
- Modeled regulatory workflows in BPMN for LEAP, a Spring Boot system digitizing SECP's compliance and approval processes
- Wrote and refined SRS documents onsite with SECP stakeholders, turning BPR requirements into structured workflow specifications
- Identified a gap in how consumers navigate SECP policies and proposed an AI chatbot using RAG over policy documents

Product Experience Analyst Sep. 2026 – Present
US Mobile (US-based MVNO on Verizon, AT&T, and T-Mobile) *Lahore, Pakistan*
- Resolve customer issues over live chat on plans, data, international roaming, and eSIM, working daily with internal admin tools

Projects

TalimBot: AI-Powered Interview Coach & Hiring Assistant for Pakistan
Python, FastAPI, LangGraph, GPT-oss-120b (Groq), Deepgram Nova-3, ElevenLabs, PostgreSQL, React (Vite)
- Developed a voice-first AI interview coaching and hiring assistant; one of about 1,100 teams invited to present on-site at the Alibaba Cloud AI Hackathon Pakistan 2026
- Orchestrated stateful agent workflows with LangGraph, which calls GPT-oss-120b on Groq for question generation, answer evaluation, and scoring; built the voice loop with MediaRecorder audio capture, Deepgram Nova-3 transcription, and ElevenLabs text-to-speech
- Architected an async FastAPI backend with auth, interview, history, CV/JD document parsing, and proctor routers; secured sessions with JWT and bcrypt, and stored users, interviews, Q&A, and scores in PostgreSQL

RAG-Based Research Paper Assistant with Transformer Models
PyTorch, BERT, GPT, T5, Qwen2.5-7B-Instruct, ChromaDB, LangChain, scikit-learn, NumPy, pandas
- Built a semantic search and RAG pipeline with Qwen2.5-7B-Instruct, overlapping fixed-size chunking (800 characters, 150 overlap), and ChromaDB vector indexing for citation-grounded question answering
- Applied 8-bit quantization and MD5-based incremental indexing, and constrained answers strictly to retrieved context to limit hallucinations
- Implemented three transformer architectures on 117K ML-ArXiv papers: encoder-only (abstract classification, 81.4% accuracy), decoder-only (text generation), and encoder-decoder (abstractive summarization)

HealthAtlas (FYP): AI-Powered Prescription Digitization Platform
Python, FastAPI, Spring Boot, React, PostgreSQL, AWS RDS, Docker, YOLO, TrOCR, Flan-T5
- Architected a multi-tier platform with a Spring Boot backend, React frontend, and FastAPI ML microservices running a sequential AI pipeline: YOLO11 field detection, TrOCR text recognition, Flan-T5 XL text correction, and RxNorm normalization
- Automated prescription digitization end to end, flagging any field where model confidence falls below 90% so a human reviews only the uncertain results

EmotiBot: Multimodal Emotion-Aware Chatbot
PyTorch, Flan-T5, LoRA, Vision Transformer, Gradio
- Fine-tuned Flan-T5 with LoRA for empathetic dialogue generation, and fused Vision Transformer facial emotion recognition with text sentiment for multimodal alignment
- Evaluated emotion classification with F1 and added confidence estimation to flag mismatches between the visual and text modalities

Technical Skills

Languages: Python (primary), SQL, JavaScript, Java, C++
Agents & LLMs: LangGraph, LangChain, agent workflows, LLM APIs (Groq), prompt design, Voice agents (Deepgram STT, ElevenLabs TTS)
RAG & Fine-tuning: RAG, ChromaDB (vector database), Semantic Search, HuggingFace, LoRA (Flan-T5), Quantization (8-bit); studied SFT and RLHF
ML: PyTorch, TensorFlow, scikit-learn, NumPy, pandas, NLP, Computer Vision, Model Evaluation
Backend & Databases: FastAPI (async), Django REST Framework, Spring Boot, REST APIs, JWT; PostgreSQL, SQLite
DevOps & Cloud: Git, Docker, Docker Compose, GitHub Actions, CI/CD, Linux, AWS (EC2, RDS, S3)

Achievements

Alibaba Cloud AI Hackathon Pakistan 2026 2026
TalimBot: AI-Powered Interview Coach & Hiring Assistant for Pakistan
- Selected among 1100 projects from 16,000+ applications